

CS4430 – Network Security Assignment 6: Network Telescope Analysis

Zubair Al-Zubi (5342007)

June 13, 2025

This assignment report presents an analysis of network traffic captured by a telescope operating in a certain enterprise. The main goals are to spot scanning activity, find the top scanner IPs, and see which protocols are used most. The analysis was done using Python scripts on the provided PCAP files.

1 Top Scanners

Table 1 shows the top 10 most active scanning sources by packet count. These IPs generated over 9% of the total traffic, with IP 185.242.226.50 scanning over 188,000 packets, which could indicate automated scanning tools or botnet activity.

IP Address	Packets	Share
185.242.226.50	188,293	2.32%
185.242.226.46	105,263	1.30%
45.146.224.132	74,096	0.91%
10.14.176.3	72,407	0.89%
10.241.128.246	72,326	0.89%
80.75.212.37	59,683	0.74%
183.136.225.42	56,189	0.69%
212.70.149.138	54,256	0.67%
118.123.105.92	51,992	0.64%
218.92.0.99	49,420	0.61%

Table 1: Top 10 scanner IP addresses by packet count

2 Top Target Ports

Table 2 lists the top destination ports targeted by TCP and UDP scanners. Most TCP activity focused on ports commonly associated with remote access or web services (such as 22 (SSH), 3389 (RDP), 80 (HTTP)), while UDP scans primarily targeted infrastructure and communication services (such as 514 (Syslog), 53 (DNS), 5060 (SIP)). This indicates that the attackers are probing for both administrative access points and commonly exposed lightweight services.

Port	Packets
22	174,938
3389	101,839
80	90,764
8088	82,746
8080	80,344
8728	76,159
5555	69,330
443	53,354
8122	51,620
12103	50,686

(a) TCP Scans

Port	Packets
514	144,733
53	91,129
5060	63,392
1194	40,973
32410	27,957
123	23,558
55001	13,926
1900	13,211
6881	13,009
5353	11,856

(b) UDP Scans

Table 2: Top 10 destination ports for TCP and UDP scans

3 Protocol Breakdown

Figure 1 shows a pie chart of the protocol breakdown of the captured traffic. TCP dominates with 7,404,342 packets (91.4%) versus UDP's 697,351 packets (8.6%). This reflects typical scanning patterns targeting web and remote access services.

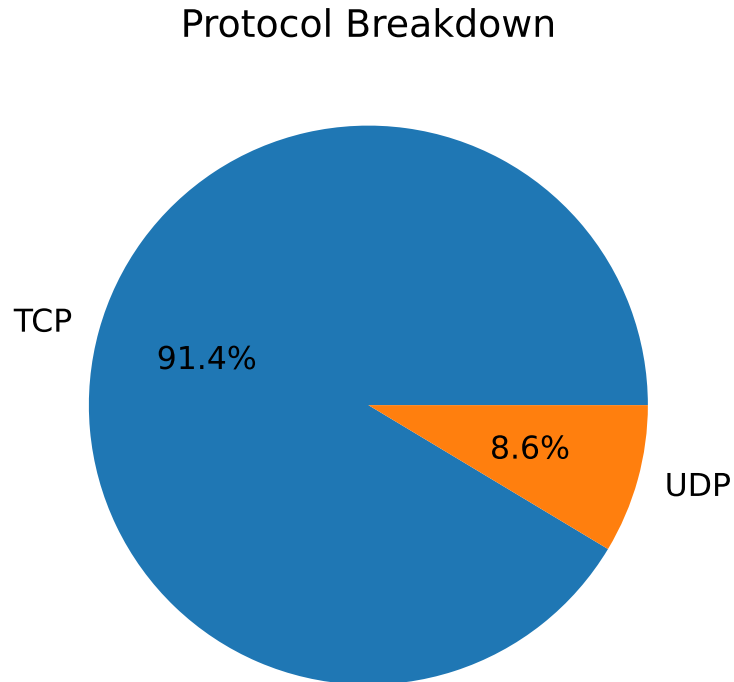


Figure 1: Protocol breakdown showing TCP vs UDP packet counts in the dataset

4 Heavy Hitter Scanner

Table 3 shows the identified scanner with the highest number of packets, which is IP address 185.242.226.50 and it has sent over 188,000 TCP SYN and UDP packets, targeting 49,853 unique IP addresses and 4 different ports. This indicates a broad scanning strategy, likely automated. The IP geolocates to the Netherlands (country code NL).

Field	Value
Scanner IP	185.242.226.50
Number of packets	188,293
Number of destination IPs	49,853
Number of ports	4
Country	NL

Table 3: Profile of the heavy-hitter scanner

5 Visualization

5.1 Time Series

Figure 2 shows the number of scanning packets observed over time, grouped into 5-minute bins. The traffic starts at a peak of nearly 1.5 million packets and gradually decreases before showing a slight rise again near the end.

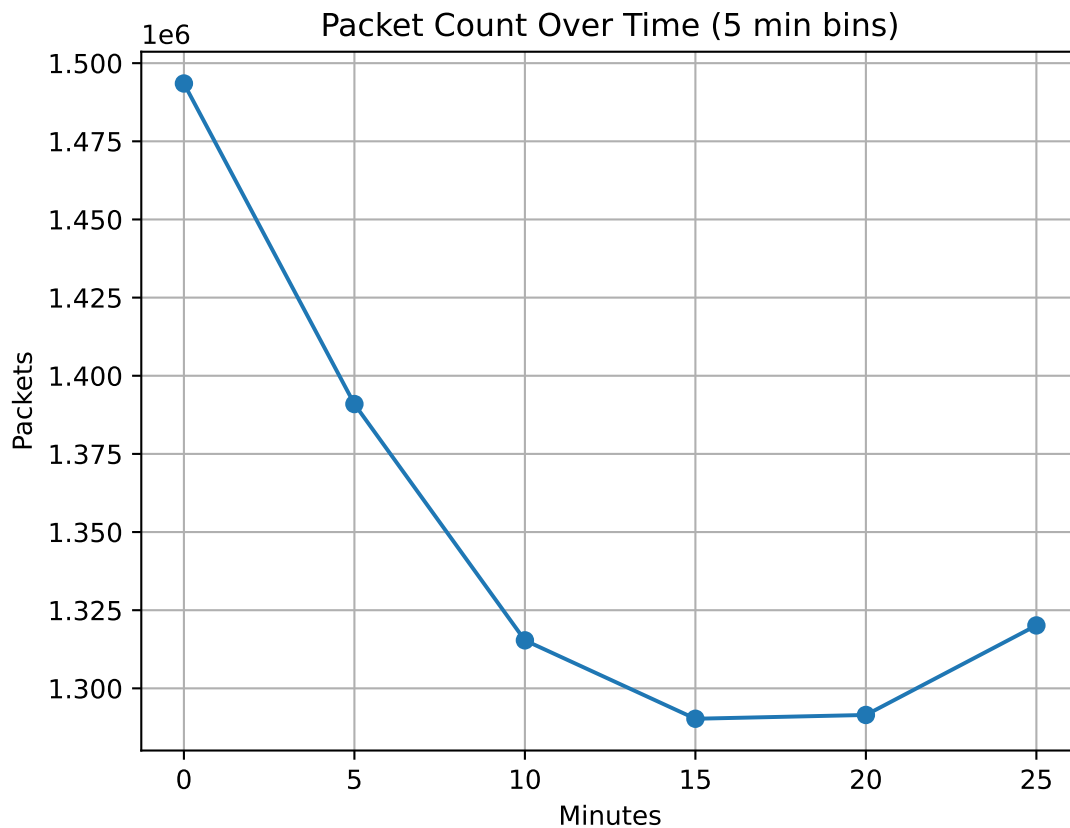


Figure 2: Packet count over time in 5-minute intervals

5.2 Port Distribution

Figure 3 shows the top 10 most scanned TCP and UDP destination ports. TCP scans target services such as SSH (22), RDP (3389), and HTTP (80), while UDP scans focus on Syslog (514), DNS (53), and SIP (5060).

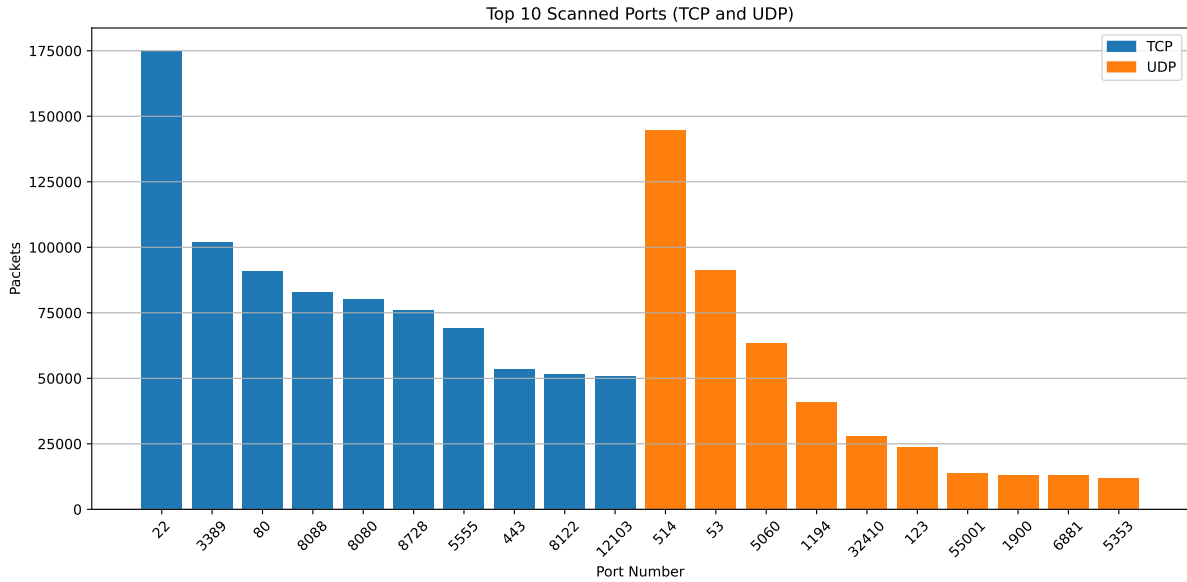


Figure 3: Top 10 scanned TCP and UDP ports

5.3 Scanner Scope

Figure 4 shows the number of unique destination IP addresses scanned by the top 10 scanner IPs. The most aggressive scanner scanned nearly 50,000 IPs.

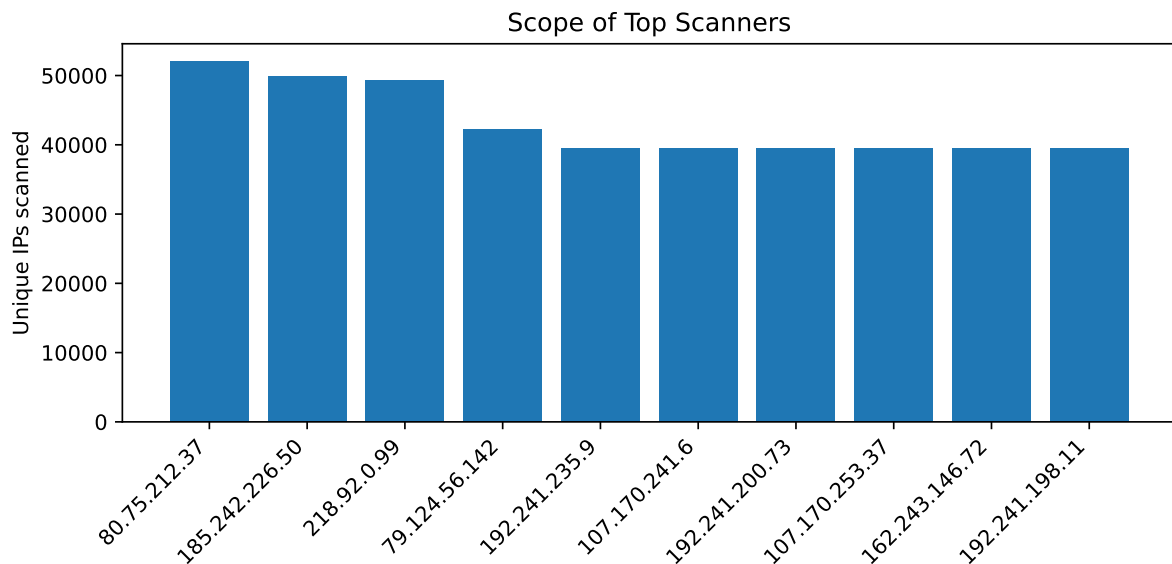


Figure 4: Number of unique IPs scanned by the top 10 scanners

6 Interpretation and Reasoning

6.1 Pattern Recognition

Based on the data and the plots, **Figure 3** reveals that the scanners appear to be mainly targeting well-known ports (such as 22, 3389, 80 for TCP and 514, 53 for UDP), which indicates service-specific reconnaissance rather than sweeping broad port ranges. Furthermore, IP 185.242.226.50 targeted nearly 50,000 unique IP addresses while only scanning four ports (**Figure 5**), which indicates a focused scanning strategy.

The overall scanning activity appears steady and persistent rather than bursty, as seen in **Figure 2**, which shows relatively consistent traffic across the capture period. This suggests automated scanning tools operating continuously.

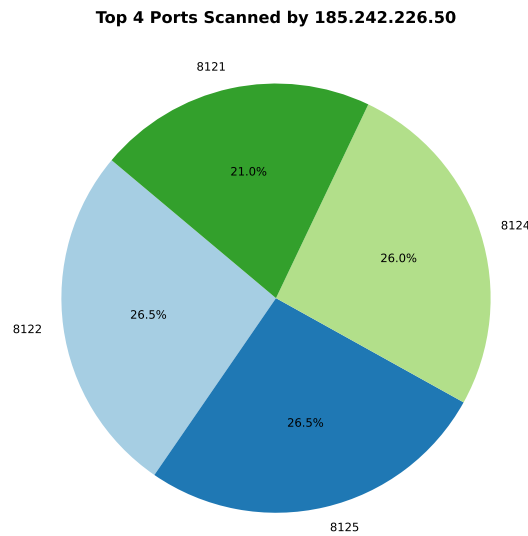


Figure 5: Port targeting by the top scanner IP

6.2 Threat Assessment

Focusing on scanner 185.242.226.50, it sent over 188,000 packets to nearly 50,000 unique IP addresses while only targeting four ports (**Figure 6**). This definitely suggests a horizontal scanning strategy, where a single service (e.g., SSH on port 22) is probed across a large address space. Thus, this scanner exhibits mass scanner characteristics.

Such scanners pose high risks by identifying open services for later exploitation, brute-force attacks, or malware delivery.

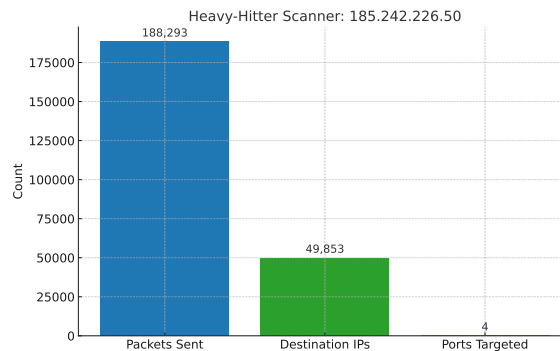


Figure 6: Heavy-hitter scanner activity: packets sent, IPs scanned, and ports targeted